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O Level

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ENVIRONMENTAL MANAGEMENT **5014/12**

Paper 1 **May/June 2016**

MARK SCHEME

Maximum Mark: 120

Published

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Section A

Question	Answer	Marks
1(a)(i)	line at 76 or 85; shading correct with ref. to key;	2
1(a)(ii)	87;	1
1(a)(iii)	*any 2 of:* few narrow valleys for dam building; few powerful rivers; few rivers with steep courses; climate too dry / dry season; plenty of other cheaper sources; HEP expensive; developing country / lack of finance; AVP;	2
1(b)	*increase in energy demand* industrial development/increase in wealth of population increases demand for electricity in homes / increased population; *increase in CO_2 emissions* reduction in use of fossil fuels/use of alternative energy sources / environmental laws enacted;	2
1(c)	*any 3 of:* vegetation cleared / loss of habitats; waste heaps / scenic beauty lost; pit head buildings / scenic beauty lost; (but allow scenic beauty lost only once) subsidence hollows; slimes; water pollution from waste washed into water courses; railways / roads; air pollution from transporting coal away; staff quarters / houses / settlement;	3

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Question	Answer	Marks
2(a)	A; D; B; E; C; all correct = [4], 3 correct = [3], 2 correct = [2], 1 correct = [1]	**4**
2(b)(i)	*any 2 of:* for agriculture / farming / arable / pastoral / animals; for irrigation; for HEP;	**2**
2(b)(ii)	*any 1 of:* poor quality; polluted / contaminated; acidic / saline; too deep underground; low population such as a mountainous area; low population in very cold area / water in form of ice/snow;	**1**
2(c)(i)	salt removed from water / soil;	**1**
2(c)(ii)	*any 2 of:* expensive; the water has a poor taste; many areas are away from coasts / sea / no access to salty water; needs much energy; often not environmentally friendly process / polluting;	**2**

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Question	Answer	Marks
3(a)(i)	line at 190 mm;	1
3(a)(ii)	all year/every month; more December to May than June to November / trend of decrease and then increase / owtte;	2
3(a)(iii)	*advantages* enough warmth for crops to grow all year; enough rain for crop growth all year / irrigation; crops grow rapidly in hot, wet conditions; double cropping is possible; abundant rain for HEP; rain all year for reliable HEP; sufficient rain for domestic / industrial use; AVP; *disadvantages* heavy rain can damage the crop; risk of flooding; rain leaches nutrients from the soils; heavy rain washes soil away / rain causes soil erosion; heavy rain turns earth roads muddy / disrupts transport; hot wet conditions / high humidity hard to work in / live in; AVP; *at least one advantage and one disadvantage for maximum mark*	3
3(b)(i)	*any 2 of:* cold; frozen ground / snow / ice; plants cannot take up water / photosynthesise for many months; short growing season / 3 months or less	2
3(b)(ii)	glasshouse / greenhouse / under glass; heated / artificial light / hydroponic / details of hydroponics / watering / irrigation;	2

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Question	Answer	Marks
4(a)	*any 3 of:* grass; short; clumps / tussocks; scrub; taiga; conical / coniferous trees; dense in background; scattered / sparse / lack of vegetation in the middle of the photograph;	**3**
4(b)(i)	*any 2 of:* gullies / dissected / owtte; development for second mark; steep slopes; development for second mark; soil eroded / removed so crops can't grow; bare / little vegetation for animals to graze;	**2**
4(b)(ii)	*any 1 of:* clearance of forest / lack of vegetation; overgrazing; run-off from heavy rain / snow melt; water run-off down the slope;	**1**
4(c)(i)	processes **changing** land in which plants can grow to land where they cannot / from productive to non-productive land / owtte;	**1**
4(c)(ii)	*any 3 of:* maintain plant / vegetation cover / prevent deforestation / afforestation; limit stock numbers rotational grazing; crop rotation / use fertiliser; controlled irrigation; introduce birth control / keep family sizes small; farmer training for sustainable land use; avoid intensive farming; plant trees as windbreaks; AVP;	**3**

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Section B

Question	Answer	Marks
5(a)(i)	660–680; 2011;	**2**
5(a)(ii)	1983;	**1**
5(a)(iii)	18 years;	**1**
5(a)(iv)	flooding;	**1**
5(b)(i)	(fruit / veg need a lot of water) so close to the river for irrigation / fertile soil close to river; extensive cattle farming has less need of water / less need of fertile soil;	**2**
5(b)(ii)	*any 2 of:* (many years of) low rainfall / below average rainfall; so grass / fodder doesn't grow much / dies back; reduced food for cattle / cattle may die; lack of water for cattle to drink; farmers lose money / go bankrupt / forced to sell cattle;	**2**
5(b)(iii)	*any 4 of:* heavy rain; so ground becomes saturated; leading to surface runoff; following (long) dry period; so would be limited vegetation; to hold soil together; when rain falls it washes soil away;	**4**

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Question	Answer	Marks
5(b)(iv)	*any 4 of:* nitrates and / or phosphates; washed into water sources; nitrates in drinking water harm human health; in rivers / lakes cause algae to increase rapidly; bacterial decomposition of algae when they die; (leading to) oxygen deficiency; and so eutrophication; leading to death of fish / invertebrates;	**4**
5(b)(v)	*any 3 of:* educating farmers; reducing fertiliser / pesticide use; using natural / organic fertilisers / such as manure / compost; using natural bio-controls for pests; less intensive fruit growing; reduce stocking levels on cattle farms; mixed cropping; crop rotation; store water for irrigation trickle drip irrigation; AVP;	**3**
5(c)(i)	period of abnormally low rainfall;	**1**
5(c)(ii)	Europe;	**1**

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Question	Answer	Marks
5(c)(iii)	mainly in the tropics; except for China / India; east side of continents; in a band through north Africa; Oceania; South / south east Asia; eastern South America; *max 2 on named locations*	**3**
5(c)(iv)	removal of vegetation / desertification; climate change / global warming;	**2**
5(d)	*any 3 of:* drought means vegetation dries out; accidental burning by farmers; climate change causing loss of habitats / lack of water; expanding human activities / deforestation; pollution of environments;	**3**
5(e)	*any 4 of:* weakening of trade winds; reversal of warm / equatorial current; to flow eastwards; increase in rainfall; so no upwelling of cold water; so lack of nutrients at surface; causing decline in numbers of marine creatures;	**4**

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Question	Answer	Marks
5(f)	*Indicative content* droughts tend to last longer and affect larger areas than floods or cyclones; floods and cyclones cause more damage to property than droughts; all can lead to soil erosion and loss of crops / animals; cyclones and floods are short-term and may require emergency rescue, shelter, food, etc.; effects of drought, being longer term, can be planned for, but can cause far more deaths than the others if no food aid, etc.; most environments recover from such disasters; it will depend on the severity of each disaster as to environmental effects, though flooding and cyclones more likely to result in pollutants being washed into water courses or the sea; droughts less environmentally damaging; Level 3 5–6 marks must communicate the features of the climatic hazards. Answers the question with detailed consideration of climatic hazards. Must look at both sides of the argument, so must be some discussion of floods and/or droughts and their relative impacts. Level 2 3–4 marks considers both sides (i.e. other climatic hazards besides cyclones) with brief details OR detailed consideration of one climatic hazard (probably cyclones) Level 1 1–2 marks basic descriptive points with little or no reasoning. May just be a list of impacts of cyclones and an agreement without mention of other climatic hazards. no response or no creditable response, 0.	**6**

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Question	Answer	Marks
6(a)(i)	crust; mantle; core; all 3 correct = [2], 1 or 2 correct = [1]	**2**
6(a)(ii)	*any 3 of:* two plates moving towards each other; both are continental plates; collision cause plates to buckle; creating folds and mountains / fold mountains;	**3**
6(a)(iii)	*any 2 of:* no oceanic plate; so no subduction; so no melting of plate for magma;	**2**
6(b)(i)	constructive (divergent);	**1**
6(b)(ii)	two arrows pointing away from central rift; fault labelled at central rift;	**2**
6(c)(i)	south / southwest of Iceland / on the Atlantic mid-oceanic ridge;	**1**
6(c)(ii)	igneous;	**1**
6(d)(i)	21 / 22;	**1**
6(d)(ii)	*any 4 of:* no plants in 1965; increased up to 1975–8; then remained constant until 1985; constant again 1987 to 1989; then a rapid increase; up to 55 / 56 plant species in 2000; accept other valid descriptions;	**4**

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Question	Answer	Marks
6(d)(iii)	*any 3 of:* sudden increase; transported seeds to Surtsey; in droppings / feathers; droppings provide nutrients for plant growth;	**3**
6(d)(iv)	*any 2 of:* change in the types of plant species (that occupy a given area through time); from bare ground to climax vegetation; can use descriptive example such as bare ground e.g. moss/lichen/trees /forest;;	**2**
6(e)(i)	*any 4 of:* cold water pumped down well; passes through cracks / joints / fissures in the rock; which heats it; hot water rises (under pressure); at surface (drop in pressure) turns water to steam; turns turbines / generator;	**4**
6(e)(ii)	has hot rocks close to the surface (or similar) / volcanic;	**1**
6(e)(iii)	yes or clearly implied; available to future generations /lasts forever; Earth's heat / water not used-up or can be reused; OR max 2 if 'no' given as answer AND reasons as 50 year max life for power plant at that location; if no control of cold water pumping;	**3**

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Question	Answer	Marks
6(e)(iv)	*any 4 of:* burning fossil fuels produces carbon dioxide / carbon monoxide; carbon dioxide is a greenhouse gas; responsible for enhanced global warming; max 1 on consequences of global warming; burning produces nitrogen / sulfur oxides; which cause acid rain; burning produces soot / particulates; which cause smog / toxic gases / health problems; max 1 for impacts of extraction if well explained;	**4**
6(f)	*Indicative content* advantages such as geothermal, fertile soils, minerals, tourist destination and therefore source of income; disadvantages such as risks from lava, lahars, ash clouds, pyroclastics, etc.; may also discuss little choice as nowhere to move to, tradition, etc., but these can only be peripheral to the argument; Level 3 5–6 marks must communicate the hazardous / beneficial features of volcanic eruptions / regions; must reach a conclusion having covered both advantages and disadvantages with developed arguments / explanations; may be more detailed on one side than the other; Level 2 3–4 marks must communicate the hazardous and/or beneficial features of volcanic eruptions / regions. covers both advantages and disadvantages with brief arguments / explanations. OR one sided looking at either advantages or disadvantages with developed arguments / explanations. Level 1 1–2 marks basic descriptive points with little or no reasoning. May just be a list of for or against. no response or no creditable response, 0.	**6**

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